

1

Introduction

Learning Objectives

After reading this chapter you should be able to

- LO 1.1 Define thermodynamics and its application in various devices
- LO1.2 Differentiate work-producing and work-consuming systems
- LO1.3 Discuss the two points of view for the analysis of thermodynamics
- LO1.4 Analyze thermodynamic system and its types
- LO1.5 Explain the concept of: State, Property, Process and Cycle
- LO1.6 Elaborate on homogeneous and heterogeneous system
- LO1.7 Describe thermodynamic Equilibrium and its satisfying conditions
- LO1.8 Explain quasi-static process related to a reversible process
- LO1.9 Discuss the concept of Continuum
- LO1.10 Explain why thermodynamics is a misnomer?
- LO1.11 List basic units adopted in SI system and different quantities derived out of them
- LO1.12 Define pressure and various pressure measuring devices
- LO1.13 Describe the journey of evolution of engineering thermodynamics

Thermodynamics is the science of energy transfer and its effect on the physical properties of substances. It is based upon observations of common experiences which have been formulated into thermodynamic laws. These laws govern the principles of energy conversion. The applications of the thermodynamic laws and principles are found in all fields of energy technology, notably in steam and nuclear power plants, internal combustion engines, gas turbines, air conditioning, refrigeration, gas dynamics, jet propulsion, compressors, chemical process plants and direct energy conversion devices.

One may be curious to know about the founders of engineering thermodynamics. A list of individuals whose lives were vital to the founding of the subject is given below with some information in regard to their contributions.

LO 1.1

Define thermodynamics and its application in various devices

1.6 MACROSCOPIC VERSUS MICROSCOPIC VIEWPOINT

There are two points of view from which the behaviour of matter can be studied: the macroscopic and the microscopic. In the macroscopic approach, a certain quantity of matter is considered, without the events occurring at the molecular level being taken into account. From the microscopic point of view, matter is composed of myriads of molecules. If it is a gas, each molecule at a given instant has a certain position, velocity and energy, and for each molecule these change very frequently as a result of collisions. The behaviour of the gas is described by summing up the behaviour of each molecule. Such a study is made in microscopic or statistical thermodynamics. Macroscopic thermodynamics is only concerned with the effects of the action of many molecules, and these effects can be perceived by human senses. For example, the macroscopic quantity, pressure, is the average rate of change of momentum due to which all the molecular collisions made on a unit area. The effects of pressure can be felt. The macroscopic point of view is not concerned with the action of individual molecules, and the force on a given unit area can be measured by using, e.g. a pressure gauge. These macroscopic observations are completely independent of the assumptions regarding the nature of matter. All the results of classical or macroscopic thermodynamics can, however, be derived from the microscopic and statistical study of matter.

LO 1.3

Discuss the two points of view for the analysis of thermodynamics

1.7 THERMODYNAMIC SYSTEM AND CONTROL VOLUME

A thermodynamic system is defined as a quantity of matter or a region in space upon which attention is concentrated in the analysis of a problem. Everything external to the system is called the surroundings or the environment. The system is separated from the surroundings by the system boundary (Fig. 1.7). The boundary may be either fixed or moving. A system and its surroundings together comprise a universe.

LO 1.4

Analyze thermodynamic system and its types

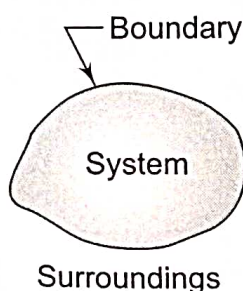


Fig. 1.7 A thermodynamic system

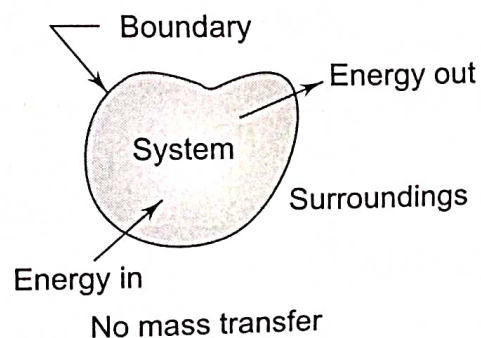


Fig. 1.8 A closed system

There are three classes of systems: (a) closed system, (b) open system and (c) isolated system. The closed system (Fig. 1.8) is a system of fixed mass. There is no mass transfer across the system boundary. There may be energy transfer into or out of the system. A certain quantity of fluid in a cylinder bounded by a piston constitutes a closed system. The open system (Fig. 1.9) is one in which matter crosses the boundary of the system. There may be energy transfer also. Most of the engineering devices are generally open systems, e.g. an air compressor in which air enters at low pressure and leaves at high pressure and there are energy transfers across the system boundary. The isolated system (Fig. 1.10) is one in which there is no interaction between the system and the surroundings. It is of fixed mass and energy, and there is no mass or energy transfer across the system boundary.

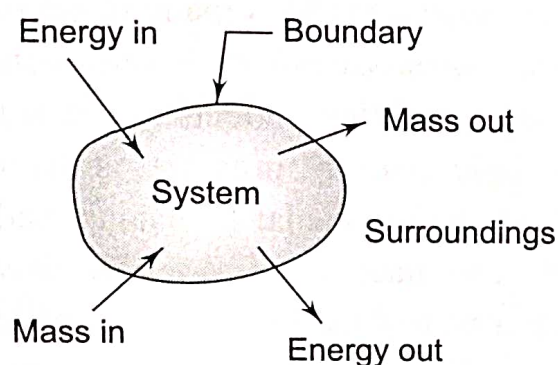


Fig. 1.9 An open system

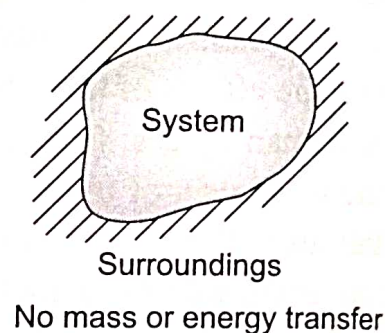


Fig. 1.10 An isolated system

If a system is defined as a certain quantity of matter, then the system contains the same matter and there can be no transfer of mass across its boundary. However, if a system is defined as a region of space within a prescribed boundary, then matter can cross the system boundary. While the former is called a closed system, the latter is an open system.

For thermodynamic analysis of an open system, such as an air compressor (Fig. 1.11), attention is focused on a certain volume in space surrounding the compressor, known as the *control volume*, bounded by a surface called the *control surface*. Matter as well as energy crosses the control surface.

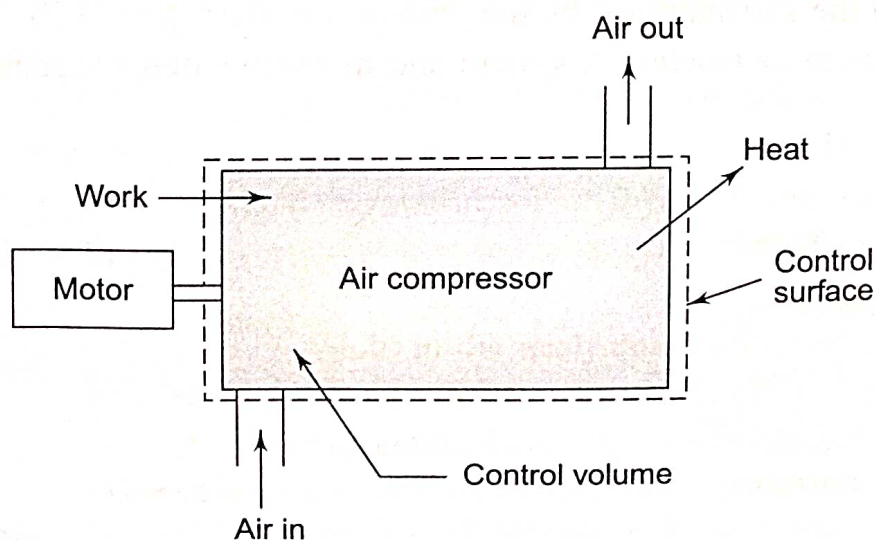


Fig. 1.11 Control volume and control surface

A closed system is a system closed to matter flow, though its volume can change against a flexible boundary. When there is matter flow, then the system is considered to be a volume of fixed identity, the control volume. There is thus no difference between an open system and a control volume.

1.8 THERMODYNAMIC PROPERTIES, PROCESSES AND CYCLES

LO 1.5

Explain the concept of: State, Property, Process and Cycle

Every system has certain characteristics by which its physical conditions may be described, e.g. volume, temperature, pressure and so on. Such characteristics are called *properties* of the system. These are all *macroscopic* in nature. When all the *properties* of a system have *definite* values, the system is said to exist at a definite *state*. Properties are the coordinates to describe the state of a system. They are the state variables of the system. Any operation in which one or more of the properties of a system change is called a *change of state*. The *succession of states passed through during a change of state* is called the *path* of the change of state. When the path is completely specified, the change of state is called a *process*, e.g. a constant pressure process. A *thermodynamic cycle* is defined as a series of state changes such that the final state is identical with the initial state (Fig. 1.12).

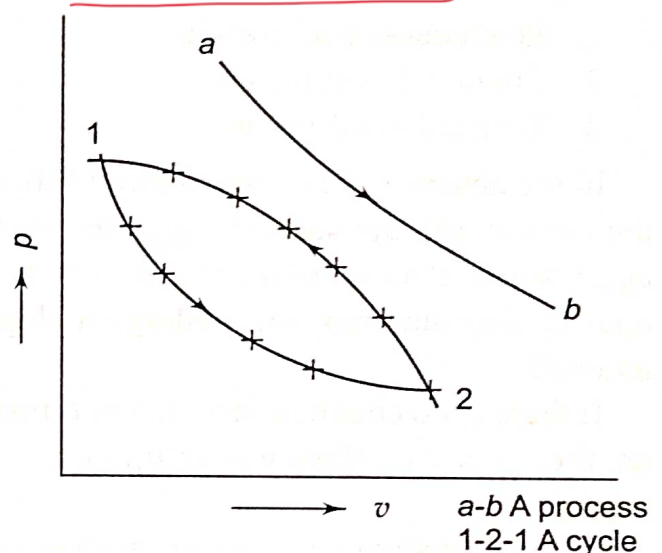


Fig. 1.12 A process and a cycle

Properties may be of two types.

Intensive properties are independent of the mass in the system, e.g. pressure, temperature, and so on. *Extensive properties* are related to mass, e.g. volume, energy and so on. If mass is increased, the values of the extensive properties also increase. Specific extensive properties, i.e. extensive properties per unit mass, are intensive properties, e.g. specific volume, specific energy and so on.

1.9 HOMOGENEOUS AND HETEROGENEOUS SYSTEMS

LO 1.6

Elaborate on homogeneous and heterogeneous system

A quantity of matter *homogeneous throughout in chemical composition and physical structure* is called a *phase*. Every substance can exist in any one of the three phases, viz. solid, liquid and gas. A system consisting of a single phase is called a *homogeneous system*, while a system consisting of more than one phase is known as a *heterogeneous system*.

1.11 QUASI-STATIC PROCESS

Let us consider a system of gas contained in a cylinder (Fig. 1.13). The system initially is in an equilibrium state, represented by the properties p_1 , v_1 and t_1 . The weight on the piston just balances the upward force exerted by the gas. If the weight is removed, there will be an unbalanced force between the system and the surroundings, and under gas pressure, the piston will move up till it hits the stops. The system again comes to an equilibrium state, being described by the properties p_2 , v_2 and t_2 . But the intermediate states passed through by the system are non-equilibrium states which cannot be described by thermodynamic coordinates.

LO 1.8

Explain quasi-static process related to a reversible process

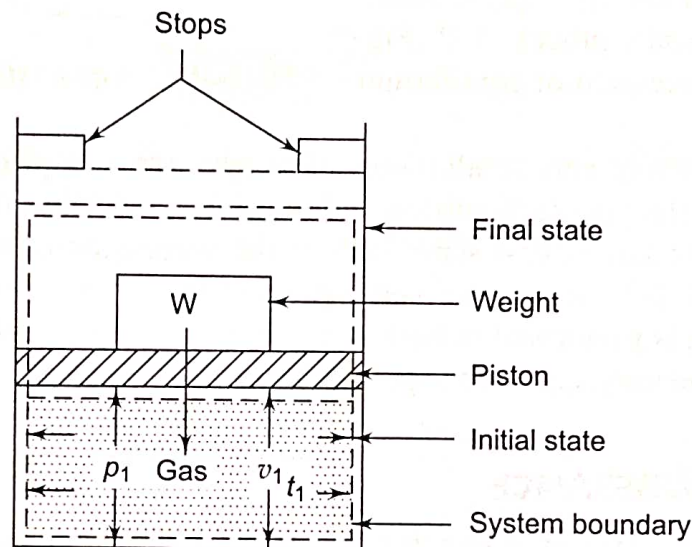


Fig. 1.13 Transition between two equilibrium states by an unbalanced force

Figure 1.14 shows points 1 and 2 as the initial and final equilibrium states joined by a dotted line, which has got no meaning otherwise. Now if the single weight on the piston is made up of many very small pieces of weights (Fig. 1.15), and these weights

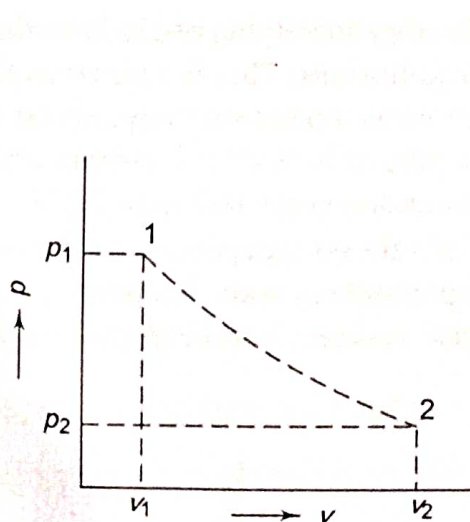


Fig. 1.14 Plot representing the transition between two equilibrium states

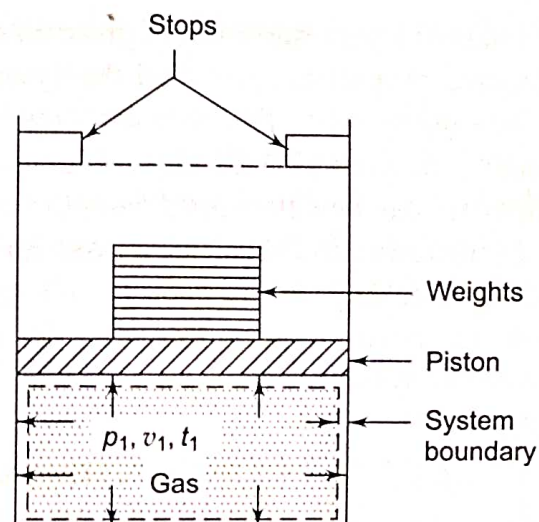


Fig. 1.15 Infinitely slow transition of a system by infinitesimal force

are removed one by one very slowly from the top of the piston, at any instant of the upward travel of the piston, if the gas system is isolated, the departure of the state of the system from the thermodynamic equilibrium state will be infinitesimally small. So every state passed through by the system will be an equilibrium state. Such a process, which is at a locus of all the equilibrium points passed through by the system, is known as a *quasi-static process* (Fig. 1.16), 'quasi' meaning 'almost'. Infinite slowness is the characteristic feature of a quasi-static process. A quasi-static process 1–2 (Fig. 1.16) is thus a succession of equilibrium states.

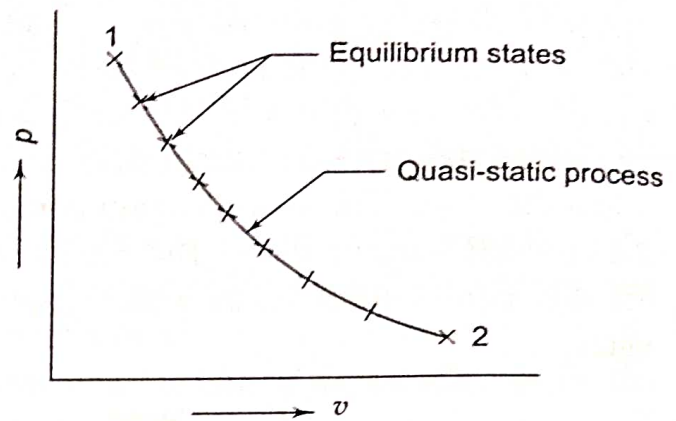


Fig. 1.16 A quasi-static process

If the same slices or very small pieces of weights are now placed slowly one by one on the top of the piston, the piston will move down slowly from state 2 to state 1 by following the same quasi-static path of the succession of equilibrium states (Figs. 1.15 and 1.16). So a quasi-static process is also a *reversible process*. A reversible process is performed in both ways from 1 to 2 and from 2 to 1 involving *infinite time* in executing each of the processes.

1.12 PURE SUBSTANCE

A pure substance is defined as one that is homogeneous and invariable in chemical composition throughout its mass. The relative proportions of the chemical elements constituting the substance are also constant. Atmospheric air, steam-water mixture and combustion products of a fuel are regarded as pure substances. But the mixture of air and liquid air is not a pure substance, since the relative proportions of oxygen and nitrogen differ in gas and liquid phases in equilibrium.

1.12.1 State Postulate

The state of a pure substance of given mass can be fixed by specifying two independent intensive properties, provided the system is in equilibrium. This is known as the '*two-property rule*'. The state can thus be represented as a point on thermodynamic property diagrams. Once any two properties of a pure substance are known, other properties can be determined from the available thermodynamic relations.

Temperature and specific volume, for example, are always independent properties, and together they can fix the state of a simple compressible system. Temperature and pressure, however, are independent for single-phase systems, but are dependent for two-phase systems.

1.13 CONCEPT OF CONTINUUM

From the macroscopic viewpoint, we are always concerned with volumes which are very large compared to molecular dimensions. Even a very small volume of a system is

LO 1.9
Discuss the concept
of Continuum

matter as a continuous medium rather than discrete particles

assumed to contain a large number of molecules so that statistical averaging is meaningful and a property value can be assigned to it. Disregarding the behaviour of individual molecules, matter is here treated as continuous. Let us consider the mass δm in a volume δV surrounding the point P (Fig. 1.17). The ratio $\delta m/\delta V$ is the average mass density of the system within the volume δV . We suppose that at first δV is rather large, and is subsequently shrunk about the point P. If we plot $\delta m/\delta V$ against δV , the average density tends to approach an asymptote as δV increases (Fig. 1.18). However, when δV becomes so small as to contain relatively few molecules, the average density fluctuates substantially with time as molecules pass into and out of the volume in random motion, and so it is impossible to speak of a definite value of $\delta m/\delta V$. The smallest volume which may be regarded as continuous is $\delta V'$. The density ρ of the system at a point is thus defined as

$$\rho = \lim_{\delta V \rightarrow \delta V'} \frac{\delta m}{\delta V} \quad (1.1)$$

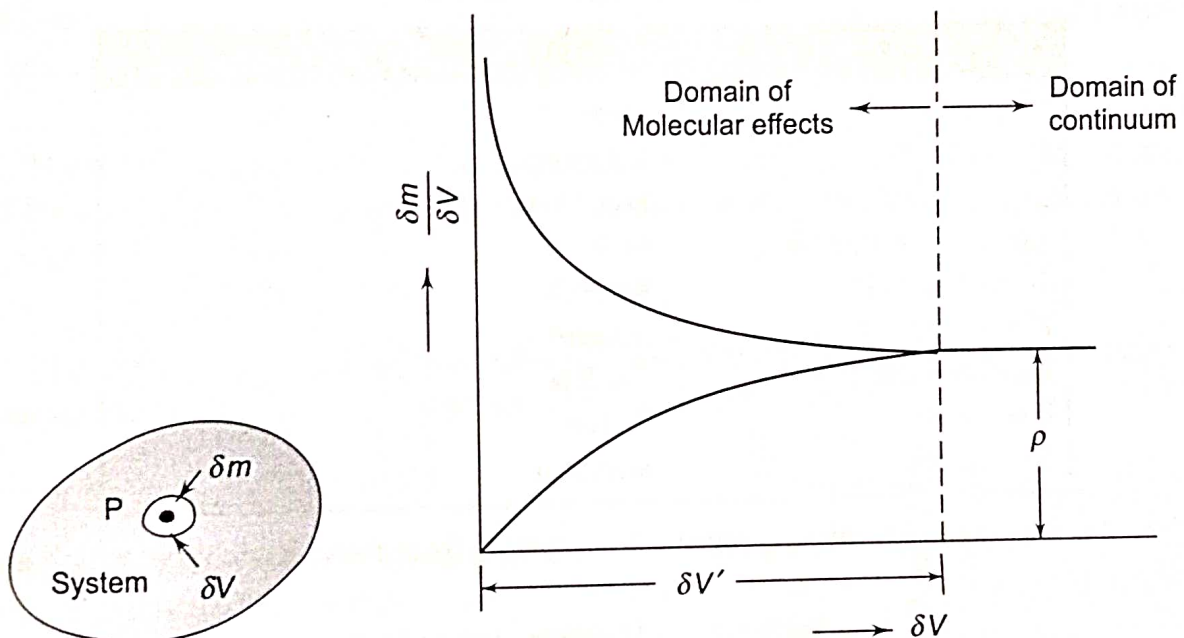


Fig. 1.17

Fig. 1.18 Definition of the macroscopic property, density

Similarly, the fluid velocity at a point P is defined as the instantaneous velocity of the centre of gravity of the smallest continuous volume $\delta V'$.

The concept of continuum loses validity when the mean free path of the molecules approaches the order of magnitude of the dimensions of the vessel, as, for instance, in highly rarefied gases encountered in high vacuum technology, in rocket flights at high altitudes and in electron tubes. In most engineering applications, however, the assumption of a continuum is valid and convenient, and goes hand in hand with the macroscopic point of view.

1.14 THERMOSTATICS

The science of thermodynamics deals with systems existing in thermodynamic equilibrium states which are specified by properties. Infinitely slow quasi-static processes executed by systems are only meaningful in thermodynamic plots.

LO 1.10

Explain why thermodynamics is a misnomer?

16 Engineering Thermodynamics

The name 'thermodynamics' is thus said to be a misnomer, since it does not deal with the dynamics of heat, which is non-quasi-static. The name 'thermostatics' then seems to be more appropriate. However, most of the real processes are dynamic and non-quasi-static, although the initial and final states of the system might be in equilibrium. Such processes can be successfully dealt with by the subject. Hence, the term 'thermodynamics' is not inappropriate.

Temperature

Learning Objectives

After reading this chapter you should be able to

- LO 2.1 State Zeroth law of thermodynamics
- LO 2.2 Discuss the thermometric property of various thermometers and methods used for temperature measurement
- LO 2.3 Compare standard parameters of various thermometers
- LO 2.4 Describe ideal gas equation of state and gas thermometers
- LO 2.5 Explain how to use ideal gas concept to obtain ideal gas temperature scale
- LO 2.6 Relate Celsius temperature scale with ideal gas scale
- LO 2.7 Examine the thermometric property utilized in electrical resistance thermometer
- LO 2.8 Understand the principle of working of thermocouple
- LO 2.9 Discuss about international practical temperature scale and its need

2.1 ZEROTH LAW OF THERMODYNAMICS

The property which distinguishes thermodynamics from other sciences is temperature. One might say that temperature bears as important a relation to thermodynamics as force does to statics or velocity does to dynamics. Temperature is associated with the ability to distinguish hot from cold. When two bodies at different temperatures are brought into contact, after sometime they attain a common temperature and are then said to exist in thermal equilibrium.

When a body A is in thermal equilibrium with a body B, and also separately with a body C, then bodies B and C will be in thermal equilibrium with each other.

This is known as the *zeroth law of thermodynamics*. It is the basis of temperature measurement.

In order to obtain a quantitative measure of temperature, a reference body is used, and a certain physical characteristic of this body which changes with temperature is selected. The changes in the selected characteristic may be taken as an indication

LO 2.1

State Zeroth law of thermodynamics

of changes in temperature. The selected characteristic is called the thermometric property, and the reference body which is used in the determination of temperature is called the thermometer. A very common thermometer consists of a small amount of mercury in an evacuated capillary tube. In this case the extension of the mercury in the tube is used as the thermometric property.

There are five different kinds of thermometer, each with its own thermometric property, as shown in Table 2.1.

2.2 MEASUREMENT OF TEMPERATURE—THE REFERENCE POINTS

The temperature of a system is a property that determines whether or not a system is in thermal equilibrium with other systems. If a body is at, say, 70°C, it will be 70°C, whether measured by a mercury-in-glass thermometer, resistance thermometer or constant volume gas thermometer. If X is the thermometric property, let us arbitrarily choose for the temperature common to the thermometer and to all systems in thermal equilibrium with it the following linear function of X :

LO 2.2

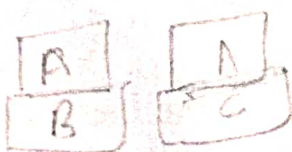
Discuss the thermometric property of various thermometers and methods used for temperature measurement

Table 2.1 Thermometers and thermometric properties

Thermometer	Thermometric property	Symbol
1. Constant volume gas thermometer	Pressure	p
2. Constant pressure gas thermometer	Volume	V
3. Electrical resistance thermometer	Resistance	R
4. Thermocouple	Thermal e.m.f.	ϵ
5. Mercury-in-glass thermometer	Length	L

$\theta(X) = aX$, where a is an arbitrary constant.

If X_1 corresponds to $\theta(X_1)$, then X_2 will correspond to



that is

$$\theta(X_2) = \frac{\theta(X_1)}{X_1} \cdot X_2 \quad (2.1)$$

Two temperatures on the linear X scale are to each other as the ratio of the corresponding X 's.

2.10 INTERNATIONAL PRACTICAL TEMPERATURE SCALE

An international temperature scale was adopted at the Seventh General Conference on Weights and Measures held in 1927. It was not to replace the Celsius or ideal gas scales, but to provide a scale that could be easily and rapidly used to calibrate the scientific and industrial instruments. Slight refinements were incorporated into the scale in revisions adopted in 1948, 1954, 1960 and 1968. The international practical scale agrees with the Celsius scale at the defining fixed points listed in Table 2.2. The temperature interval from the oxygen point to the gold point is divided into three main parts, as given below.

LO 2.9

Discuss about
international
practical
temperature scale
and its need

1. From 0 to 660°C A platinum resistance thermometer with a platinum wire whose diameter must lie between 0.05 and 0.20 mm is used, and the temperature is given by the equation:

$$R = R_0 (1 + At + Bt^2)$$

where the constants R_0 , A and B are computed by measurements at the ice point, steam point and sulphur point.

Table 2.2 Temperatures of fixed points

	Temperature °C
Normal boiling point of oxygen	− 182.97
Triple point of water (standard)	+ 0.01
Normal boiling point of water	100.00
Normal boiling point of sulphur	444.60
(Normal melting point of zinc-suggested as an alternative to the sulphur point)	419.50
Normal melting point of antimony	630.50
Normal melting point of silver	960.80
Normal melting point of gold	1063.00

2. From −190 to 0°C The same platinum resistance thermometer is used, and the temperature is given by

$$R = R_0 [1 + At + Bt^2 + C(t - 100)t^3]$$

where R_0 , A and B are the same as before, and C is determined from a measurement at the oxygen point.

3. From 660 to 1063°C A thermocouple, one wire of which is made of platinum and the other of an alloy of 90 percent platinum and 10 percent rhodium, is used with one junction at 0°C. The temperature is given by the formula:

$$\mathcal{E} = a + bt + ct^2$$

where a , b and c are computed from measurements at the antimony point, silver point and gold point. The diameter of each wire of the thermocouple must lie between 0.35 and 0.65 mm.

An optical method is adopted for measuring temperatures higher than the gold point. The intensity of radiation of any convenient wavelength is compared with the intensity of radiation of the same wavelength emitted by a black body at the gold point. The temperature is then determined with the help of Planck's law of thermal radiation.

4. A revised international temperature Scale, ITS-90, was adapted in 1990 by the International Committee of Weights and Measures with further refinements defining more fixed points and conforming more closely the thermodynamic temperature scale based on the the second law of thermodynamics (See Chapter 6).

First Law of Thermodynamics

Learning Objectives

After reading this chapter you should be able to

- LO 4.1 Explain first law for a closed system executing a cycle
- LO 4.2 Discuss first law for a closed system in a change of state
- LO 4.3 Describe energy as a point function and a system property
- LO 4.4 Differentiate between Energy in transit and Energy in storage
- LO 4.5 Define specific heat of a substance at constant volume
- LO 4.6 Demonstrate the concept of enthalpy and its relation with heat transfer
- LO 4.7 Define specific heat of a substance at constant pressure
- LO 4.8 Analyze the amount of total energy in an isolated system
- LO 4.9 Discuss possibility of perpetual motion machine of the first kind (PMM1)

4.1 FIRST LAW FOR A CLOSED SYSTEM UNDERGOING A CYCLE

The transfer of heat and the performance of work may both cause the same effect in a system. Heat and work are different forms of the same entity, called energy, which is conserved.

Energy which enters a system as heat may leave the system as work, or energy which enters the system as work may leave as heat.

Let us consider a closed system which consists of a known mass of water, m , contained in an adiabatic vessel having a thermometer and a paddle wheel, as shown in Fig. 4.1. Let a certain amount of work W_{1-2} be done upon the system by the paddle wheel. The quantity of work can be measured by the fall of weight which drives the paddle wheel through a pulley. The system was initially at temperature t_1 , the same as that of atmosphere, and after work transfer let the temperature rise to t_2 . The pressure is always 1 atm. The process 1–2 undergone by the system is shown in Fig. 4.2 in

LO 4.1

Explain first law for a closed system executing a cycle

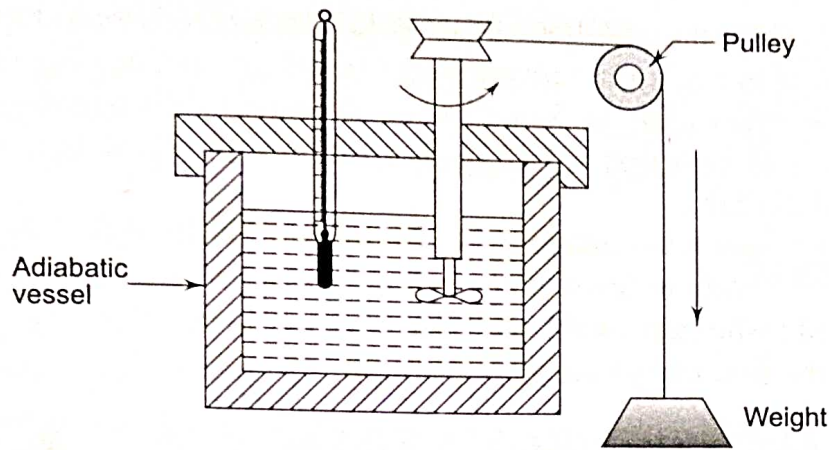


Fig. 4.1 Adiabatic work

generalised thermodynamic coordinates X , Y . Let the insulation now be removed. The system and the surroundings interact by heat transfer till the system returns to the original temperature t_1 , attaining the condition of thermal equilibrium with the atmosphere. The amount of heat transfer Q_{2-1} from the system during this process, 2-1, shown in Fig. 4.2, can be estimated from $Q_{2-1} = mc_p(t_2 - t_1)$. The system thus executes a cycle, which consists of a definite amount of work input W_{1-2} to the system followed by the transfer of an amount of heat Q_{2-1} from the system. It has been found that this W_{1-2} is always proportional to the heat Q_{2-1} , and the constant of proportionality is called the Joule's equivalent or the *mechanical equivalent of heat*. In the simple example given here, there are only two energy transfer quantities as the system performs a thermodynamic cycle. If the cycle involves many more heat and work quantities, the same result will be found. Expressed algebraically,

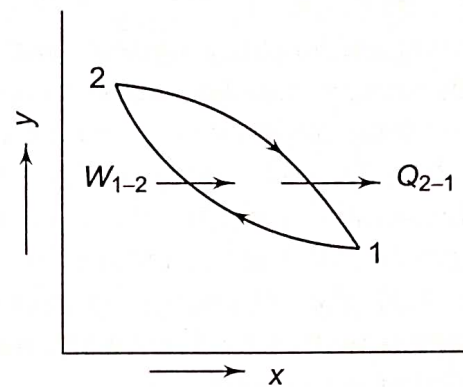


Fig. 4.2 Cycle completed by a system with two energy interactions: adiabatic work transfer W_{1-2} followed by heat transfer Q_{2-1}

$$(\Sigma W)_{\text{cycle}} = J (\Sigma Q)_{\text{cycle}} \quad (4.1)$$

where J is the Joule's equivalent. This is also expressed in the form

$$\oint dW = J \oint dQ$$

where the symbol $\oint$ denotes the cyclic integral for the closed path. This is the first law for a closed system undergoing a cycle. It is accepted as a general law of nature, since no violation of it has ever been demonstrated.

In the S.I. system of units, both heat and work are measured in the derived unit of energy, the Joule. The constant of proportionality, J , is therefore unity ($J = 1 \text{ Nm/J}$).

The first law of thermodynamics owes much to J.P. Joule who, during the period 1840–1849, carried out a series of experiments to investigate the equivalence of work and heat. In one of these experiments, Joule used an apparatus similar to the one shown in Fig. 4.1. Work was transferred to the measured mass of water by means of paddle wheel driven by the falling weight. The rise in the temperature of water was recorded. Joule also used mercury as the fluid system, and later a solid system of metal blocks which absorbed work by friction when rubbed against each other. Other

experiments involved the supplying of work in an electric current. $W_{1-2} = EIt = Q_{2-1} = mc_p(t_2 - t_1)$, where E is the voltage and I is the current flowing through the wire for t seconds. In every case, he found the same ratio (J) between the amount of work and the quantity of heat that would produce identical effects in the system, where J is unity in S.I. System.

Prior to Joule, heat was considered to be an invisible fluid flowing from a body of higher calorie to a body of lower calorie, and this was known as the *caloric theory of heat*. It was Joule who first established that heat is a form of energy, and thus laid the foundation of the first law of thermodynamics.

4.2 FIRST LAW FOR A CLOSED SYSTEM UNDERGOING A CHANGE OF STATE

The expression $(\Sigma W)_{\text{cycle}} = (\Sigma Q)_{\text{cycle}}$ applies only to systems undergoing cycles, and the algebraic summation of all energy transfer across system boundaries is zero. But if a system undergoes a change of state during which both heat transfer and work transfer are involved, the *net* energy transfer will be stored or accumulated within the system. If Q is the amount of heat transferred to the system and W is the amount of work transferred from the system during the process (Fig. 4.3), the net energy transfer ($Q - W$) will be stored in the system. Energy in storage is neither heat nor work, and is given the name *internal energy* or simply, the *energy* of the system.

Therefore,

$$Q - W = \Delta E$$

where ΔE is the increase in the energy of the system

or

$$Q = \Delta E + W \quad (4.2)$$

Here Q , W and ΔE are all expressed in the same units (in joules). Energy may be stored by a system in different modes, as explained in Section 4.4.

If there are more energy transfer quantities involved in the process, as shown in Fig. 4.4, the first law gives

$$(Q_2 + Q_3 - Q_1) = \Delta E + (W_2 + W_3 - W_1 - W_4)$$

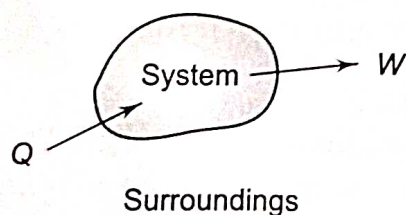


Fig. 4.3 Heat and work interactions of a system with its surroundings in a process

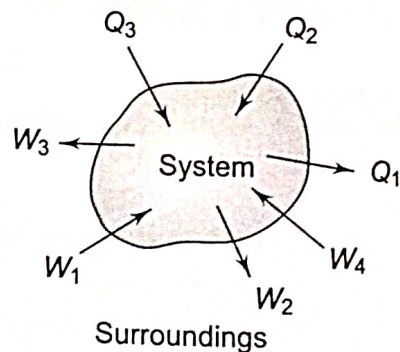


Fig. 4.4 System-surroundings interaction in a process involving many energy fluxes

Energy is thus conserved in the operation. The first law is a particular formulation of the principle of the conservation of energy. Equation (4.2) may also be considered as the definition of energy. This definition does not give an absolute value of energy E , but only the change of energy ΔE for the process. It can, however, be shown that the energy has a definite value at every state of a system and is, therefore, a property of the system.

4.3 ENERGY—A PROPERTY OF THE SYSTEM

Consider a system which changes its state from state 1 to state 2 by following the path A , and returns from state 2 to state 1 by following the path B (Fig. 4.5). So the system undergoes a cycle. Writing the first law for path A

$$Q_A = \Delta E_A + W_A \quad (4.3)$$

and for path B

$$Q_B = \Delta E_B + W_B \quad (4.4)$$

The processes A and B together constitute a cycle, for which

$$(\Sigma W)_{\text{cycle}} = (\Sigma Q)_{\text{cycle}}$$

$$\text{or } W_A + W_B = Q_A + Q_B$$

$$\text{or } Q_A - W_A = W_B - Q_B \quad (4.5)$$

From Eqs. (4.3), (4.4) and (4.5), it yields

$$\Delta E_A = -\Delta E_B \quad (4.6)$$

Similarly, had the system returned from state 2 to state 1 by following the path C instead of path B

$$\Delta E_A = -\Delta E_C \quad (4.7)$$

From Eqs. (4.6) and (4.7)

$$\Delta E_B = \Delta E_C \quad (4.8)$$

Therefore, it is seen that the change in energy between two states of a system is the same, whatever path the system may follow in undergoing that change of state. If some arbitrary value of energy is assigned to state 2, the value of energy at state 1 is fixed independent of the path the system follows. Therefore, energy has a definite value for every state of the system. Hence, it is a point function and a property of the system.

The energy E is an extensive property. The *specific energy*, $e = E/M$ (J/kg), is an intensive property.

The cyclic integral of any property is zero, because the final state is identical with the initial state. $\oint dE = 0$, $\oint dV = 0$, etc. So for a cycle, the Eq. (4.2) reduces to Eq. (4.1).

4.4 DIFFERENT FORMS OF STORED ENERGY

It was stated at the beginning that thermodynamics is the science of energy transfer and it effect the physical properties of a substance. Energy, as we know, is the capacity of doing work. In thermodynamics, energy can be in two forms: (i) Energy in transit, (ii) Energy in storage. Work and heat intractions are the forms of energy in transit, observed at the boundaries of a system. They are not the properties of a system. They are path functions, their magnitudes depending upon the path the system follows during a change of state. Energy in storage, called internal energy, is a point or state function and hence a property of a system.

LO 4.3

Describe energy as a point function and a system property

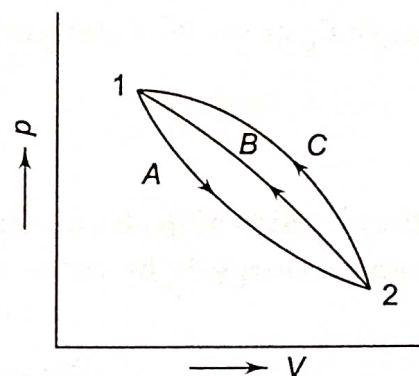


Fig. 4.5 Energy—A property of a system

LO 4.4

Differentiate between Energy in transit and Energy in storage

The symbol E refers to the total energy stored in a system. Basically there are two modes in which energy may be stored in a system:

1. Macroscopic energy mode
2. Microscopic energy mode

The macroscopic energy mode includes the macroscopic kinetic energy and potential energy of a system. Let us consider a fluid element of mass m having the centre of mass velocity or barycentric velocity $\bar{V}$ (Fig. 4.6). The macroscopic kinetic energy E_K of the fluid element by virtue of its motion is given by

$$E_K = \frac{m\bar{V}^2}{2}$$

If the elevation of the fluid element from an arbitrary datum is z , then the macroscopic potential energy E_p by virtue of its position is given by

$$E_p = mgz$$

The microscopic energy mode refers to the energy stored in the molecular and atomic structure of the system, which is called the *molecular internal energy* or *simply internal energy*, customarily denoted by the symbol U . Matter is composed of molecules. Molecules are in random thermal motion (for a gas) with an average velocity $\bar{v}$, constantly colliding with one another and with the walls (Fig. 4.6). Due to a collision, the molecules may be subjected to rotation as well as vibration. They can have translational kinetic energy, rotational kinetic energy, vibrational energy, electronic energy, chemical energy and nuclear energy (Fig. 4.7). If ϵ represents the energy of one molecule, then

$$\epsilon = \epsilon_{\text{trans}} + \epsilon_{\text{rot}} + \epsilon_{\text{vib}} + \epsilon_{\text{chem}} + \epsilon_{\text{electronic}} + \epsilon_{\text{nuclear}} \quad (4.9)$$

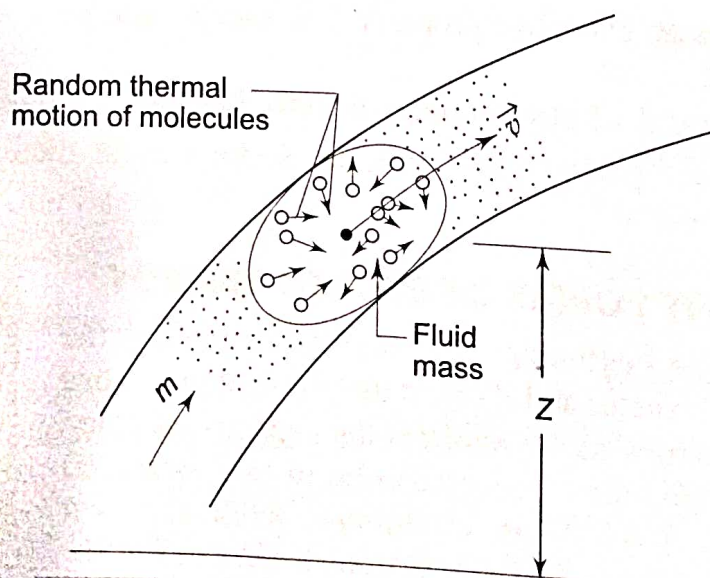


Fig. 4.6 Macroscopic and microscopic energy

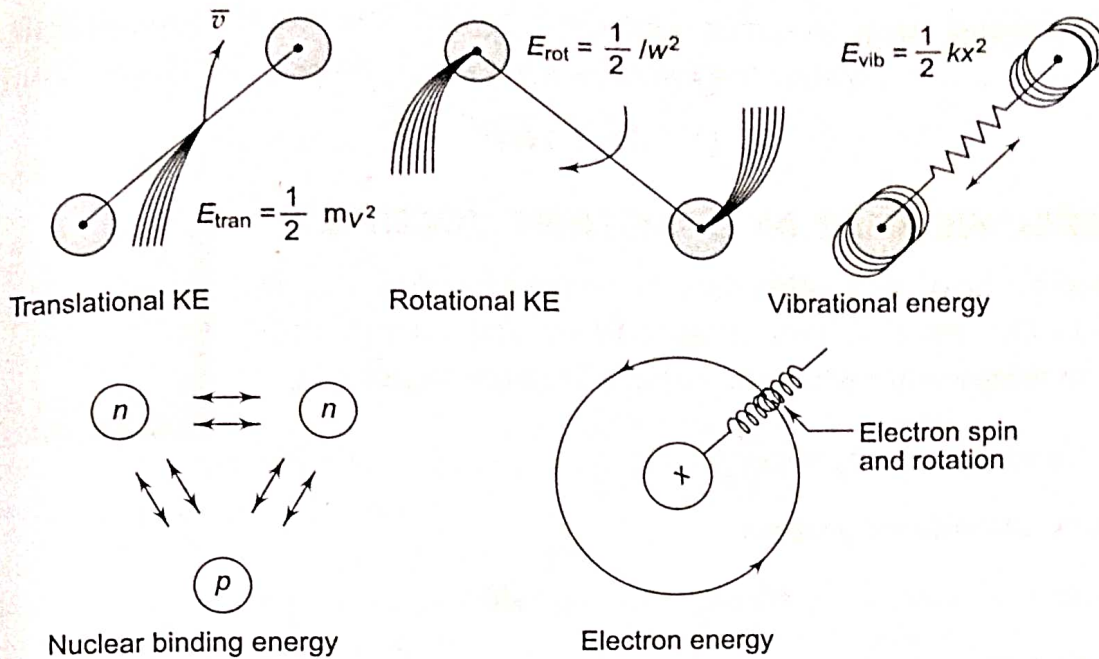


Fig. 4.7 Various components of internal energy stored in a molecule

If N is the total number of molecules in the system, then the total internal energy

$$U = N\varepsilon \quad (4.10)$$

In an ideal gas there are no intermolecular forces of attraction and repulsion, and the internal energy depends only on temperature. Thus,

$$U = f(T) \text{ only} \quad (4.11)$$

for an ideal gas.

Other forms of energy which can also be possessed by a system are magnetic energy, electrical energy and surface (tension) energy. In the absence of these forms, the total energy E of a system is given by

$$E = \underbrace{E_K + E_P}_{\text{macro}} + \underbrace{U}_{\text{micro}} \quad (4.12)$$

where E_K , E_P and U refer to the kinetic, potential and internal energy, respectively. In the absence of motion and gravity

$$E_K = 0, \quad E_P = 0$$

$$E = U$$

and Eq. (4.2) becomes

$$Q = \Delta U + W \quad (4.13)$$

U is an extensive property of the system. The specific internal energy u is equal to U/m and its unit is J/kg.

In the differential forms, Eqs. (4.2) and (4.13) become

$$dQ = dE + dW \quad (4.14)$$

$$dQ = dU + dW \quad (4.15)$$

where

$$dW = \delta W_{pdV} + dW_{\text{shaft}} + dW_{\text{electrical}} + \dots,$$

considering the different forms of work transfer which may be present. When only pdV work is present, the equations become

$$dQ = dE + pdV \quad (4.16)$$

$$dQ = dU + pdV \quad (4.17)$$

4.8 ENERGY OF AN ISOLATED SYSTEM

An isolated system is one in which there is no interaction of the system with the surroundings. For an isolated system

$$\delta Q = 0 \quad \delta W = 0.$$

The first law gives

$$dE = 0$$

or

$$E = \text{constant}$$

The energy of an isolated system is always constant.

LO4.8

Analyze the amount of total energy in an isolated system

4.9 PERPETUAL MOTION MACHINE OF THE FIRST KIND—PMM1

The first law states the general principle of the conservation of energy. *Energy is neither created nor destroyed, but only gets transformed from one form to another.* There can be no machine which would continuously supply mechanical work without some other form of energy disappearing simultaneously (Fig. 4.8). Such a fictitious machine is called a perpetual motion machine of the first kind, or in brief, PMM1. A PMM1 is thus impossible.

LO4.9

Discuss possibility of perpetual motion machine of the first kind (PMM1)

The converse of the above statement is also true, i.e. there can be no machine which would continuously consume work without some other form of energy appearing simultaneously (Fig. 4.9).

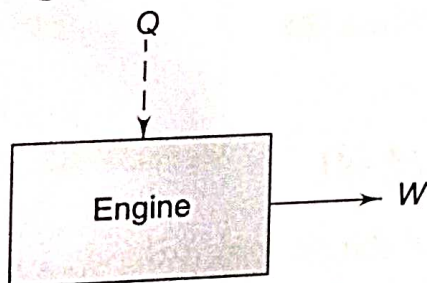


Fig. 4.8 PMM1

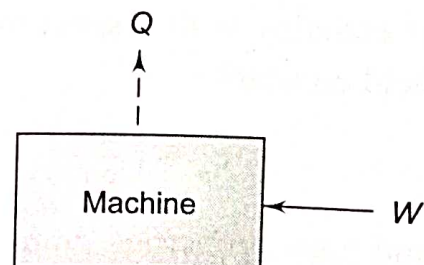


Fig. 4.9 Converse of PMM1

5

First Law Applied to Flow Processes

Learning Objectives

After reading this chapter you should be able to

- LO 5.1 Explain flow processes in thermodynamics using fluid mechanics approaches
- LO 5.2 Describe the concept of steady flow
- LO 5.3 Discuss derivation of steady flow energy equation (S.F.E.E.) on per unit mass and time basis
- LO 5.4 Discuss the application of S.F.E.E. in most common engineering devices
- LO 5.5 Formulate Euler's and Bernoulli's equation from S.F.E.E.
- LO 5.6 Define an unsteady flow process i.e. variable flow process
- LO 5.7 Analyse variable flow process using both system boundary and control volume technique
- LO 5.8 Develop an equation in a variable flow process applied for discharging and charging a tank

5.1 CONTROL VOLUME

For any system and in any process, the first law can be written as

$$Q = \Delta E + W$$

where E represents all forms of energy stored in the system.

For a pure substance,

$$E = E_K + E_P + U$$

where E_K is the K.E., E_P is the P.E. and U is the residual energy stored in the molecular structure of the substance.

$$\therefore Q = \Delta E_K + \Delta E_P + \Delta U + W$$

LO 5.1

Explain flow processes in thermodynamics using fluid mechanics approaches

(5.1)

When there is mass transfer across the system boundary, the system is called an open system. Most of the engineering devices are open systems involving the flow of fluids through them.

Equation (5.1) refers to a system having a particular mass of substance, and it is free to move from place to place.

Consider a steam turbine (Fig. 5.1) in which steam enters at a high pressure, does work upon the turbine rotor, and then leaves the turbine at low pressure through the exhaust pipe.

If a certain mass of steam is considered as the thermodynamic system, then the energy equation becomes

$$Q = \Delta E_K + \Delta E_P + \Delta U + W$$

and in order to analyse the expansion process in turbine the moving system is to be followed as it travels through the turbine, taking into account the work and heat interactions all the way through. This method of analysis is similar to that of Lagrange in fluid mechanics.

Although the system approach is quite valid, there is another approach which is found to be highly convenient. Instead of concentrating attention upon a certain quantity of fluid, which constitutes a moving system in flow processes, attention is focused upon a certain fixed region in space called a *control volume* through which the moving substance flows. This is similar to the analysis of Euler in fluid mechanics.

To distinguish the two concepts, it may be noted that while the system (closed) boundary usually changes shape, position and orientation relative to the observer, the control volume boundary remains fixed and unaltered. Again, while matter usually crosses the control volume boundary, no such flow occurs across the system boundary.

The broken line in Fig. 5.1 represents the surface of the control volume which is known as the *control surface*. This is the same as the system boundary of the open system. The method of analysis is to inspect the control surface and account for all energy quantities transferred through this surface. Since there is mass transfer across the control surface, a mass balance also has to be made. Sections 1 and 2 allow mass transfer to take place, and Q and W are the heat and work interactions, respectively.

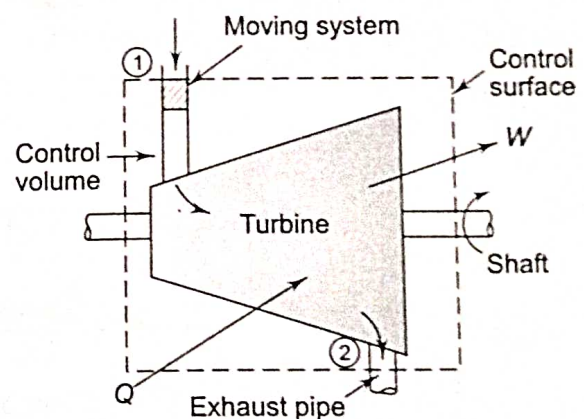


Fig. 5.1 Flow process involving work and heat interactions

5.2 STEADY FLOW PROCESS

As a fluid flows through a certain control volume, its thermodynamic properties may vary along the space coordinates as well as with time. If the rates of flow of mass and energy through the control surface change with time, the mass and energy within the control volume also would change with time.

LO 5.2

Describe the concept of steady flow

'Steady flow' means that the rates of flow of mass and energy across the control surface are constant.

In most engineering devices, there is a constant rate of flow of mass and energy through the control surface, and the control volume in course of time attains a steady state. At the steady state of a system, any thermodynamic property will have a fixed value at a particular location, and will not alter with time. Thermodynamic properties may vary along space coordinates, but do not vary with time. 'Steady state' means that the state is steady or invariant with time.

5.3 MASS BALANCE AND ENERGY BALANCE IN A SIMPLE STEADY FLOW PROCESS

In Fig. 5.2, a steady flow system has been shown in which, one stream of fluid enters and another stream leaves the control volume. There is no accumulation of mass or energy within the control volume, and the properties at any location within the control volume are steady with time.

Sections 1-1 and 2-2 indicate, respectively, the entrance and exit of the fluid across the control surface. The following quantities are defined with reference to Fig. 5.2:

A_1, A_2 —cross section of stream, m^2

w_1, w_2 —mass flow rate, kg/s

p_1, p_2 —pressure, absolute, N/m^2

$$w_1 = \frac{dm_1}{dt} = \frac{dm_1}{dt}$$

$$w_2 = \frac{dm_2}{dt} = \frac{dm_2}{dt}$$

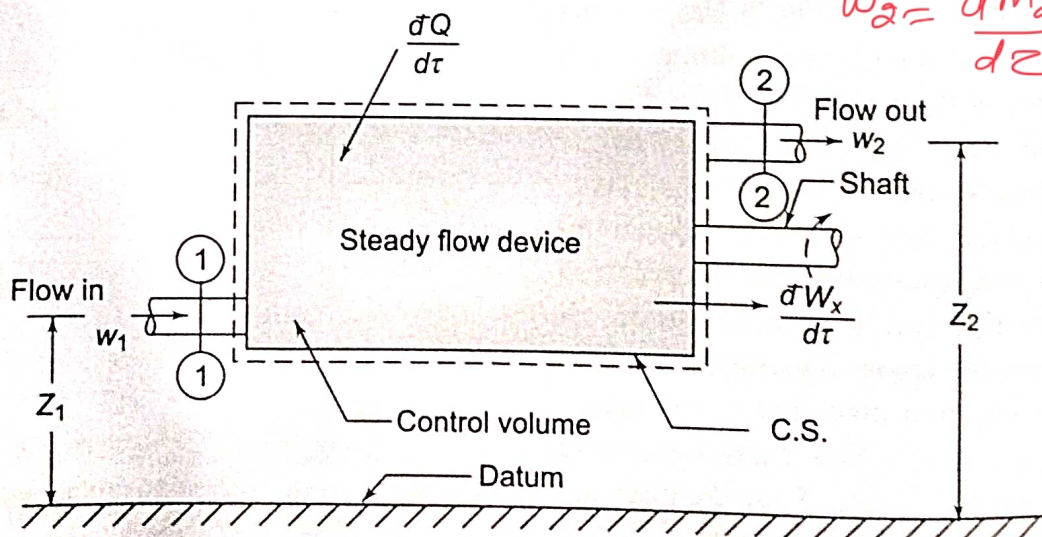


Fig. 5.2 Steady flow process

v_1, v_2 —specific volume, m^3/kg

u_1, u_2 —specific internal energy, J/kg

V_1, V_2 —velocity, m/s

Z_1, Z_2 —elevation above an arbitrary datum, m

LO 5.3

Discuss derivation of steady flow energy equation (S.F.E.E) on per unit mass and time basis

$\frac{\dot{d}Q}{d\tau}$ —net rate of heat transfer through the control surface, J/s

$\frac{\dot{d}W_x}{d\tau}$ —net rate of work transfer through the control surface, J/s

exclusive of work done at Sections 1 and 2 in transferring the fluid through the control surface.

τ —time, s.

Subscripts 1 and 2 refer to the inlet and exit sections.

5.3.1 Mass Balance

By the conservation of mass, if there is no accumulation of mass within the control volume, the mass flow rate entering must equal the mass flow rate leaving, or

$$\text{or} \quad \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad \Rightarrow \quad \frac{dm_1}{d\tau} = \frac{dm_2}{d\tau} \quad (5.2)$$

This equation is known as the equation of continuity.

5.3.2 Energy Balance

In a flow process, the work transfer may be of two types: the *external work* and the *flow work*. The external work refers to all the work transfer across the control surface other than that due to normal fluid forces. In engineering thermodynamics, the only two kinds of external work of importance are *shear* (shaft or stirring) *work* and *electrical work*. In Fig. 5.2 the only external work occurs in the form of shaft work, W_x . The flow work, as discussed in Section 3.4, is the displacement work done by the fluid of mass dm_1 at the inlet Section 1 and that of mass dm_2 at the exit Section 2, which are $(-p_1 v_1 dm_1)$ and $(+p_2 v_2 dm_2)$, respectively. Therefore, the total work transfer is given by

$$W = W_x - p_1 v_1 dm_1 + p_2 v_2 dm_2 \quad (5.3)$$

In the rate form,

$$\frac{\dot{d}W}{d\tau} = \frac{\dot{d}W_x}{d\tau} - p_1 v_1 \frac{dm_1}{d\tau} + p_2 v_2 \frac{dm_2}{d\tau}$$

$$\text{or} \quad \frac{\dot{d}W}{d\tau} = \frac{\dot{d}W_x}{d\tau} - w_1 p_1 v_1 + w_2 p_2 v_2 \quad (5.4)$$

Since there is no accumulation of energy, by the conservation of energy, the total rate of flow of all energy streams entering the control volume must equal the total rate of flow of all energy streams leaving the control volume. This may be expressed in the following equation:

$$w_1 e_1 + \frac{\dot{d}Q}{d\tau} = w_2 e_2 + \cancel{\frac{\dot{d}Q}{d\tau}} + \frac{\dot{d}W}{d\tau}$$

Substituting for $\frac{\dot{d}W}{d\tau}$ from Eq. (5.4)

$$w_1 e_1 + \frac{\dot{d}Q}{d\tau} = w_2 e_2 + \frac{\dot{d}W_x}{d\tau} - \underbrace{w_1 p_1 v_1 + w_2 p_2 v_2}_{\text{flow work}}$$

$$\therefore w_1 e_1 + w_1 p_1 v_1 + \frac{dQ}{d\tau} = w_2 e_2 + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \quad (5.5)$$

where e_1 and e_2 refer to the energy carried into or out of the control volume with unit mass of fluid.

The specific energy e is given by

$$e = e_k + e_p + u = \frac{V^2}{2} + Zg + u \quad (5.6)$$

Substituting the expression for e in Eq. (5.5)

$$\begin{aligned} & w_1 \left(\underbrace{\frac{V_1^2}{2} + Z_1 g + u_1}_{e_1} \right) + w_1 p_1 v_1 + \frac{dQ}{d\tau} \\ &= w_2 \left(\underbrace{\frac{V_2^2}{2} + Z_2 g + u_2}_{e_2} \right) + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \\ \text{or} \quad & w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ &= w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{dW_x}{d\tau} \quad (5.7) \end{aligned}$$

where $h = u + pv$.

And, since

$$w_1 = w_2, \text{ let } w = w_1 = w_2 = \frac{dm}{d\tau}$$

steady state

Dividing Equation (5.7) by $\frac{dm}{d\tau}$

$$h_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = h_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW}{dm} \quad (5.8)$$

Equations (5.7) and (5.8) are known as steady flow energy equations (S.F.E.E.), for a single stream of fluid entering and a single stream of fluid leaving the control volume. All the terms in Equation (5.8) represent energy flow per unit mass of fluid (J/kg), whereas the terms in Equation (5.7) represent energy flow per unit time (J/s). The basis of energy flow per unit mass is usually more convenient when only a single stream of fluid enters and leaves a control volume. When more than one fluid stream is involved the basis of energy flow per unit time is more convenient.

Equation (5.8) can be written in the following form:

$$Q - W_x = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + g(Z_2 - Z_1) \quad (5.9)$$

where Q and W_x refer to energy transfer per unit mass. In the differential form, the SFEE becomes

$$dQ - dW_x = dh + \bar{V}d\bar{V} + g dZ \quad (5.10)$$

When more than one stream of fluid enters or leaves the control volume (Fig. 5.3), the mass balance and energy balance for steady flow are given as follows.

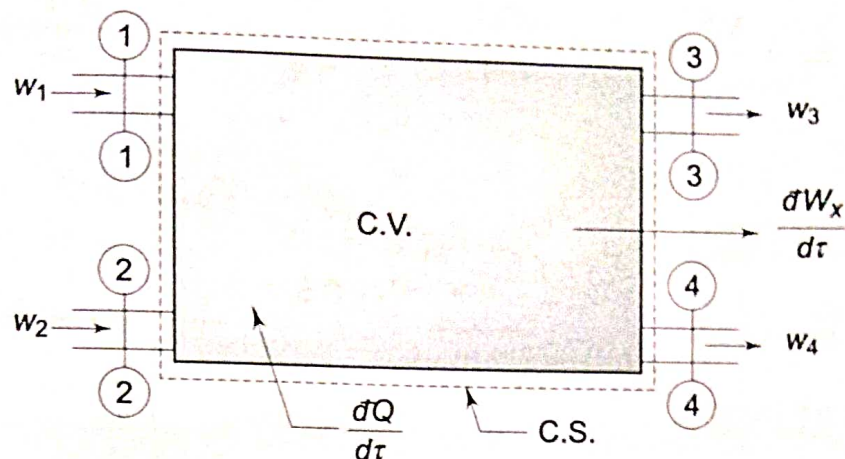


Fig. 5.3 Steady flow process involving two fluid streams at the inlet and exit of the control volume

Mass balance

$$w_1 + w_2 = w_3 + w_4 \quad (5.11)$$

$$\frac{A_1 V_1}{v_1} + \frac{A_2 V_2}{v_2} = \frac{A_3 V_3}{v_3} + \frac{A_4 V_4}{v_4} \quad (5.12)$$

Energy balance

$$\begin{aligned} & w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{dQ}{d\tau} \\ &= w_3 \left(h_3 + \frac{V_3^2}{2} + Z_3 g \right) + w_4 \left(h_4 + \frac{V_4^2}{2} + Z_4 g \right) + \frac{dW_x}{d\tau} \end{aligned} \quad (5.13)$$

The steady flow energy equation applies to a wide variety of processes like pipe line flows, heat transfer processes, mechanical power generation in engines and turbines, combustion processes and flows through nozzles and diffusers. In certain problems, some of the terms in steady flow energy equation may be negligible or zero. But it is best to write the full equation first, and then eliminate the terms which are unnecessary.

5.4 SOME EXAMPLES OF STEADY FLOW PROCESSES

The following examples illustrate the applications of the steady flow energy equation in some of the engineering systems.

5.4.1 Nozzle and Diffusor

A nozzle is a device which increases the velocity or K.E. of a fluid at the expense of its pressure drop, whereas a diffusor increases the pressure of a fluid at the expense of

LO 5.4

Discuss the application of S.F.E.E. in most common engineering devices

its K.E. Figure 5.4 shows a nozzle which is insulated. The steady flow energy equation of the control surface gives

$$h_1 + \frac{V_1^2}{2} + Z_1 g + \frac{\dot{d}Q}{dm} = h_2 + \frac{V_2^2}{2} + Z_2 g + \frac{\dot{d}W_x}{dm}$$

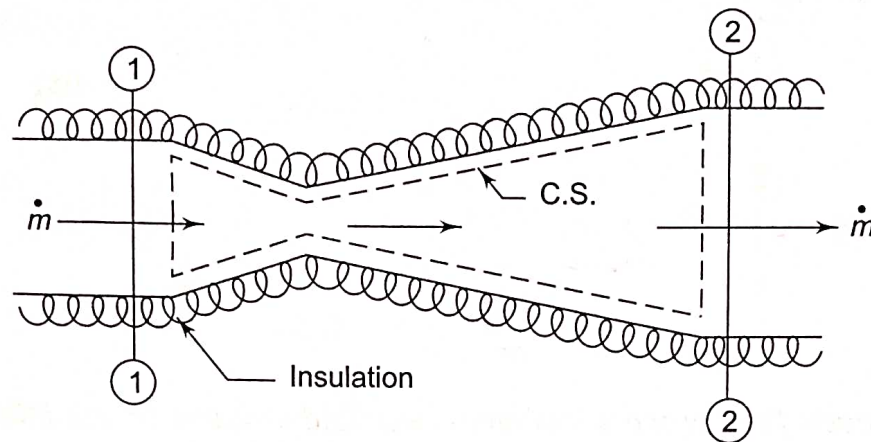


Fig. 5.4 Steady flow process involving two fluid streams at the inlet and exit of the control volume

Here, $\frac{\dot{d}Q}{dm} = 0$, $\frac{\dot{d}W_x}{dm} = 0$ and the change in potential energy is zero. The equation reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad (5.14)$$

The continuity equation gives

$$w = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad (5.15)$$

When the inlet velocity or the 'velocity of approach' V_1 is small compared to the exit velocity V_2 , Eq. (5.14) becomes

$$h_1 = h_2 + \frac{V_2^2}{2}$$

or

$$V_2 = \sqrt{2(h_1 - h_2)} \text{ m/s}$$

where $(h_1 - h_2)$ is in J/kg.

Equations (5.14) and (5.15) hold good for a diffuser as well.

5.4.2 Throttling Device

When a fluid flows through a constricted passage, like a partially opened valve, an orifice, or a porous plug, there is an appreciable drop in pressure, and the flow is said to be throttled. Figure 5.5 shows the process of throttling by a partially opened valve on a fluid flowing in an insulated pipe. In the steady-flow energy Eq. (5.8),

$$\frac{\dot{d}Q}{dm} = 0, \quad \frac{\dot{d}W_x}{dm} = 0$$

and the changes in P.E. are very small and ignored. Thus, the S.F.E.E. reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

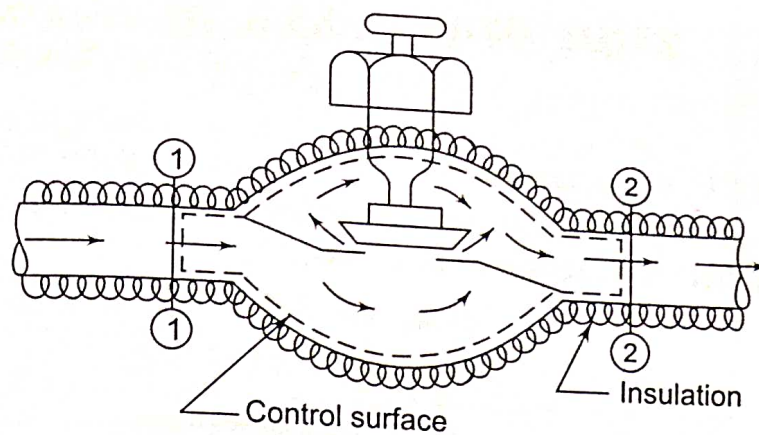


Fig. 5.5 Flow through a valve

Often the pipe velocities in throttling are so low that the K.E. terms are also negligible. So

$$h_1 = h_2 \quad (5.16)$$

or the enthalpy of the fluid before throttling is equal to the enthalpy of the fluid after throttling.

Enthalpy is the energy of a flowing fluid per unit mass. Since no energy is imparted to, or rejected from, the fluid during a throttling process, its enthalpy i.e., energy remains the same.

5.4.3 Turbine and Compressor

Turbines and engines give positive power output, whereas compressors and pumps require power input.

For a turbine (Fig. 5.6) which is well insulated, the flow velocities are often small, and the K.E. terms can be neglected. The S.F.E.E. then becomes

$$h_1 = h_2 + \frac{\delta W_x}{dm}$$

or

$$\frac{W_x}{m} = (h_1 - h_2)$$

It is seen that work is done by the fluid at the expense of its enthalpy.

Similarly, for an adiabatic pump or compressor, work is done upon the fluid and W is negative. So the S.F.E.E. becomes

$$h_1 = h_2 - \frac{W_x}{m}$$

or

$$\frac{W_x}{m} = h_2 - h_1$$

The enthalpy of the fluid increases by the amount of work input.

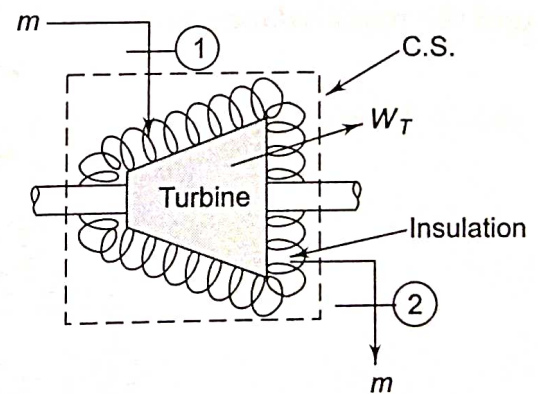


Fig. 5.6 Flow through a turbine

5.4.4 Heat Exchanger

A heat exchanger is a device in which heat is transferred from one fluid to another. Figure 5.7 shows a steam condenser, where steam condenses outside the tubes and cooling water flows through the tubes. The S.F.E.E. for the C.S. gives

$$w_c h_1 + w_s h_2 = w_c h_3 + w_s h_4$$

or

$$w_s (h_2 - h_4) = w_c (h_3 - h_1)$$

Here the K.E. and P.E. terms are considered small, there is no external work done, and energy exchange in the form of heat is confined only between the two fluids, i.e. there is no external heat interaction or heat loss.

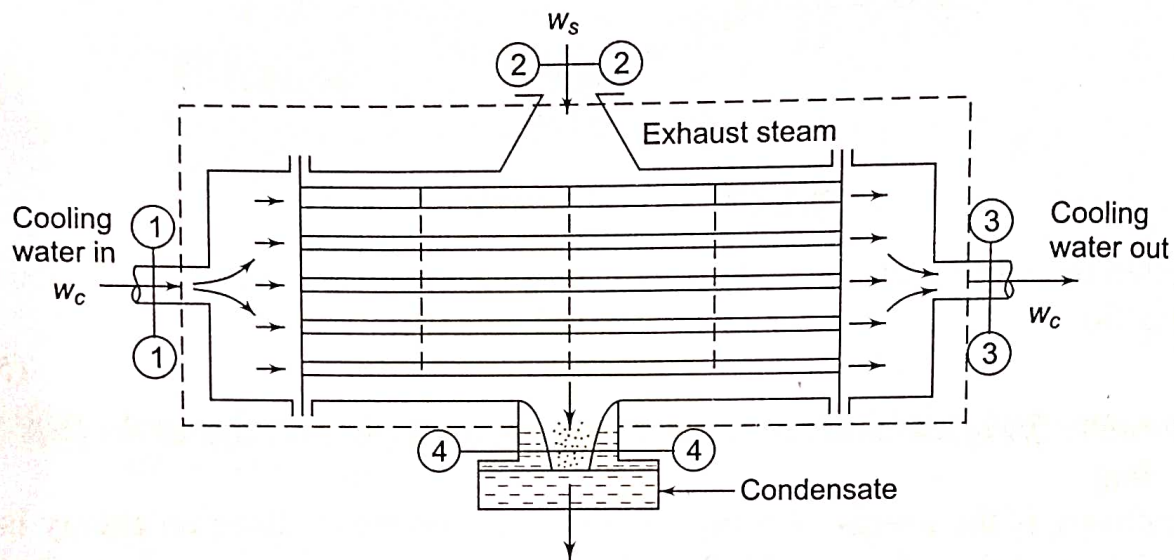


Fig. 5.7 Steam condenser

Figure 5.8 shows a steam desuperheater where the temperature of the superheated steam is reduced by spraying water. If w_1 , w_2 and w_3 are the mass flow rates of the injected water, of the steam entering, and of the steam leaving, respectively, and h_1 , h_2 and h_3 are the corresponding enthalpies, and if K.E. and P.E. terms are neglected as before, the S.F.E.E. becomes

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

and the mass balance gives

$$w_1 + w_2 = w_3$$

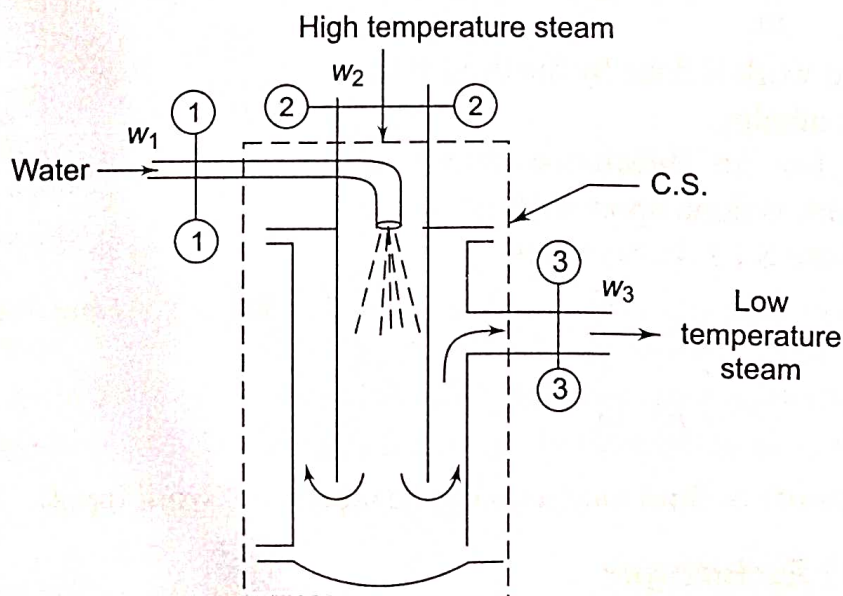


Fig. 5.8 Steam desuperheater

5.5 COMPARISON OF S.F.E.E. WITH EULER AND BERNOULLI EQUATIONS

The steady flow energy equation Eq. (5.8) can be written as

$$\frac{dQ}{dm} = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_x}{dm}$$

In the differential form the S.F.E.E. becomes

$$dQ = dh + \bar{V}d\bar{V} + g dZ + dW_x \quad (5.17)$$

where dQ and dW_x refer to unit mass of the substance. Since $h = u + pv$ and $dQ = du + pdv$ (for a quasi-static path involving only pdv -work), Eq. (5.17) can be written as

$$du + pdv = du + pdv + vdp + \bar{V}d\bar{V} + g dZ + dW_x$$

For an inviscid frictionless fluid flowing through a pipe, the equation reduces to

$$vdp + \bar{V}d\bar{V} + g dZ = 0 \quad (5.18)$$

This is the *Euler equation*. If we integrate between two Sections 1 and 2 of the pipe

$$\int_1^2 vdp + \int_1^2 \bar{V}d\bar{V} + \int_1^2 g dZ = 0$$

For an incompressible fluid, $v = \text{constant}$

$$\therefore v(p_2 - p_1) + \frac{V_2^2}{2} - \frac{V_1^2}{2} + g(Z_2 - Z_1) = 0 \quad (5.19)$$

Since the specific volume v is the reciprocal of the density ρ , we have

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + Z_1 g = \frac{p_2}{\rho} + \frac{V_2^2}{2} + Z_2 g \quad (5.20)$$

$$\text{or} \quad \frac{p}{\rho} + \frac{V^2}{2} + Zg = \text{constant} \quad (5.21)$$

This is known as the *Bernoulli equation*, which is valid for an inviscid incompressible fluid. It can also be expressed in the following form:

$$\Delta \left(pv + \frac{V^2}{2} + gZ \right) = 0 \quad (5.22)$$

where v is constant and $\Delta (...)$ means 'increase in ...'

The S.F.E.E. as given by Eq. (5.8) or Eq. (5.17) can be written with $(u + pv)$ substituted for h , as follows:

$$Q - W_x = \Delta \left(u + pv + \frac{V^2}{2} + gZ \right) \quad (5.23)$$

A comparison of Eqs. (5.22) and (5.23) shows that they have several terms in common. However, while the Bernoulli equation is restricted to frictionless incompressible fluids, the S.F.E.E. is not, and is valid for viscous compressible fluids as well. The Bernoulli equation is, therefore, a special limiting case of the more general steady flow energy equation.

LO 5.5

Formulate Euler's and Bernoulli's equation from S.F.E.E.

5.6 VARIABLE FLOW PROCESSES

Many flow processes, such as filling up and evacuating gas cylinders, are not steady. Such processes can be analysed by the control volume technique. Consider a device through which a fluid is flowing under non-steady state conditions (Fig. 5.9). The rate at which the mass of fluid within the control volume is accumulated is equal to the net rate of mass flow across the control surface, as given below.

LO 5.6
Define an unsteady flow process i.e. variable flow process

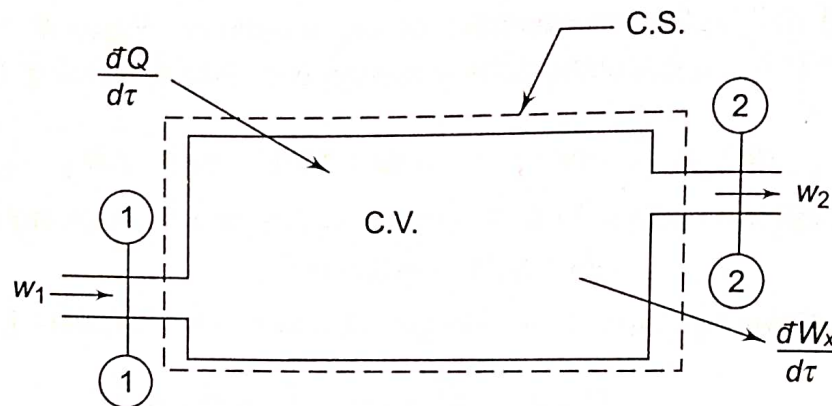


Fig. 5.9 Variable flow process

mass balance

$$\frac{dm_v}{d\tau} = w_1 - w_2 = \frac{dm_1}{d\tau} - \frac{dm_2}{d\tau} \quad (5.24)$$

where m_v is the mass of fluid within the control volume at any instant.

Over any finite period of time

$$\Delta m_v = \Delta m_1 - \Delta m_2 \quad (5.25)$$

The rate of accumulation of energy within the control volume is equal to the net rate of energy flow across the control surface. If E_v is the energy of fluid within the control volume at any instant,

Rate of energy increase = Rate of energy inflow – Rate of energy outflow

Energy balance

$$\begin{aligned} \frac{dE_v}{d\tau} = & w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ & - w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) - \frac{dW_x}{d\tau} \end{aligned} \quad (5.26)$$

Now

$$E_v = \left(U + \frac{mV^2}{2} + mgZ \right)_v$$

where m is the mass of fluid in the control volume at any instant.

$\therefore$

$$\frac{dE_v}{d\tau} = \frac{d}{d\tau} \left(U + \frac{mV^2}{2} + mgZ \right)_v$$

$$\begin{aligned}
&= \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) \frac{dm_1}{d\tau} + \frac{\delta Q}{d\tau} \\
&\quad - \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) \frac{dm_2}{d\tau} - \frac{\delta W_x}{d\tau} \quad (5.27)
\end{aligned}$$

Figure 5.10 shows all these energy flux quantities. For any finite time interval, Eq. (5.27) becomes

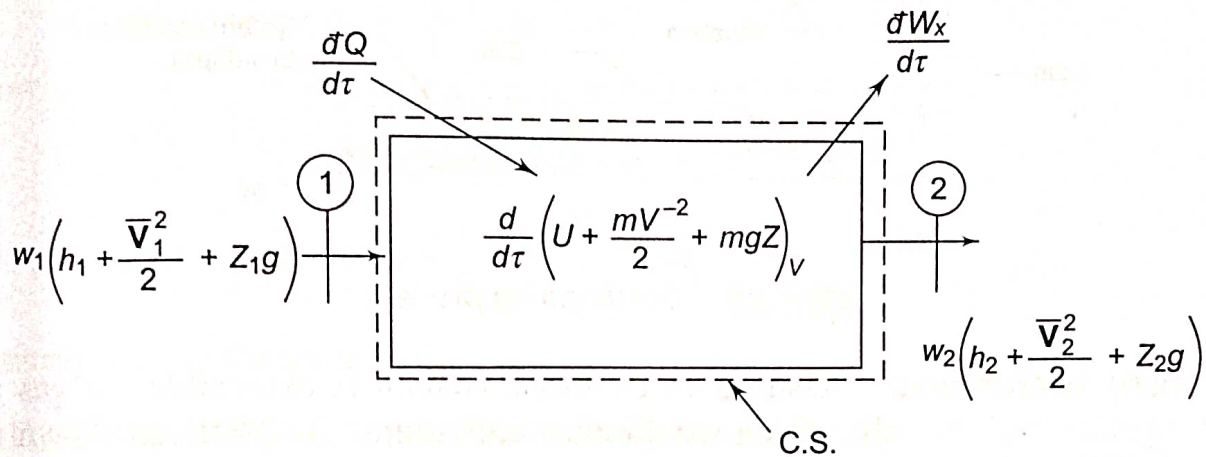


Fig. 5.10 Energy fluxes in an unsteady system

$$\begin{aligned}
\Delta E_V &= Q - W_x + \int \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) dm_1 \\
&\quad - \int \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) dm_2 \quad (5.28)
\end{aligned}$$

Equation (5.26) is the *general energy equation*. For steady flow,

$$\frac{dE_V}{d\tau} = 0,$$

and the equation reduces to Eq. (5.7). For a closed system $w_1 = 0$, $w_2 = 0$, then from Eq. (5.26),

$$\frac{dE_V}{d\tau} = \frac{\delta Q}{d\tau} - \frac{\delta W_x}{d\tau}$$

or

$$dE_V = \delta Q - \delta W_x \quad \text{or,} \quad \delta Q = \delta E + \delta W_x$$

as obtained earlier.

5.7 EXAMPLE OF A VARIABLE FLOW PROBLEM

Variable flow processes may be analysed either by the system technique or the control volume technique, illustrated as follows.

LO 5.7

Analyse variable flow process using both system boundary and control volume technique

Consider a process in which a gas bottle is filled from a pipeline (Fig. 5.11). In the beginning, the bottle contains gas of mass m_1 at state p_1, t_1, v_1, h_1 and u_1 . The valve is opened and gas flows into the bottle till the mass of gas in the bottle is m_2 at state p_2, t_2, v_2, h_2 and u_2 . The supply to the pipeline is very large so that the state of gas in the pipeline is constant at p_p, t_p, v_p, h_p, u_p and V_p .

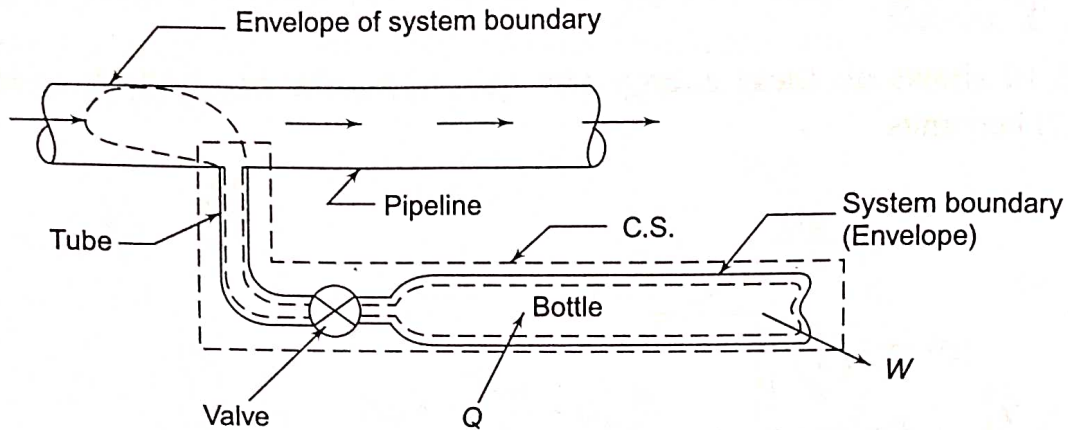


Fig. 5.11 Bottle-filling process

System technique Assume an envelope (which is extensible) of gas in the pipeline and the tube which would eventually enter the bottle as shown in Fig. 5.11.

Energy of the gas before filling

$$E_1 = m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right)$$

where $(m_2 - m_1)$ is the mass of gas in the pipeline and tube which would enter the bottle.

Energy of the gas after filling

$$E_2 = m_2 u_2$$

$$\Delta E = E_2 - E_1 = m_2 u_2 - \left[m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) \right] \quad (5.29)$$

The P.E. terms are neglected. The gas in the bottle is not in motion, and so the K.E. terms have been omitted.

Now, there is a change in the volume of gas because of the collapse of the envelope to zero volume. Then the work done

$$W = p_p (V_2 - V_1) = p_p [0 - (m_2 - m_1) v_p]$$

$$= - (m_2 - m_1) p_p v_p$$

$\therefore$ using the first law for the process

$$Q = \Delta E + W$$

$$= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) - (m_2 - m_1) p_p v_p$$

$$= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + h_p \right) \quad (5.30)$$

which gives the energy balance for the process.

Control volume technique Assume a control volume bounded by a control surface, as shown in Fig. 5.11. Applying the energy Eq. (5.27) to this case, the following energy balance may be written on a time rate basis:

$$\frac{dE_V}{d\tau} = \frac{\dot{Q}}{d\tau} + \left(h_p + \frac{V_p^2}{2} \right) \frac{dm}{d\tau}$$

Since h_p and V_p are constant, the equation is integrated to give for the total process

$$\Delta E_V = Q + \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

Now

$$\Delta E_V = U_2 - U_1 = m_2 u_2 - m_1 u_1$$

$$\therefore Q = m_2 u_2 - m_1 u_1 - \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

This equation is the same as Eq. (5.30).

5.8 DISCHARGING AND CHARGING A TANK

Let us consider a tank discharging a fluid into a supply line (Fig. 5.12). Since $\delta W_x = 0$ and $dm_{in} = 0$, applying first law to the control volume,

$$dU_V = \delta Q + \left(h + \frac{V^2}{2} + gz \right) dm_{out} \quad (5.31)$$

Assuming K.E. and P.E. of the fluid to be small and $\delta Q = 0$

$$\begin{aligned} d(mu) &= hdm \\ mdu + udm &= udm + p\bar{v} dm \end{aligned}$$

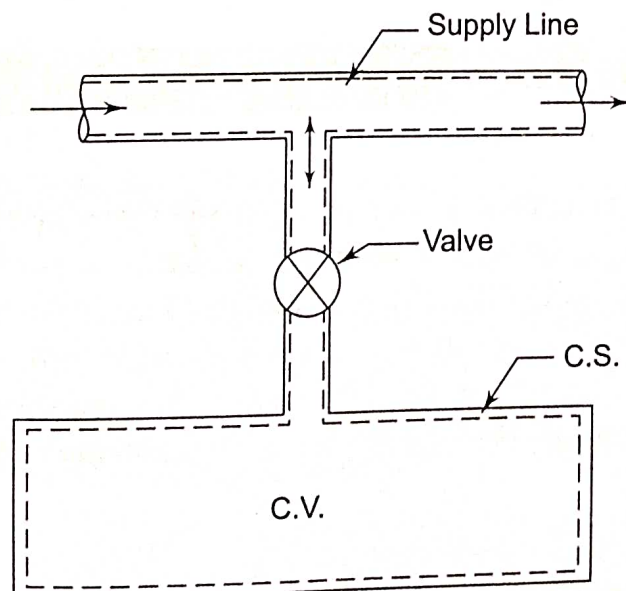


Fig. 5.12 Charging and discharging a tank

LO 5.8

Develop an equation in a variable flow process applied for discharging and charging a tank

$$\frac{dm}{m} = \frac{du}{pv} \quad (5.32)$$

Again the total volume $V = vm = \text{const.}$

$$vdm + m dv = 0$$

$$\text{or} \quad \frac{dm}{m} = -\frac{dv}{v} \quad (5.33)$$

From Eqs. (5.32) and (5.33),

$$\frac{du}{pv} = -\frac{dv}{v}$$

$$d(u + pv) = 0$$

$$\text{or} \quad \delta Q = 0$$

which shows that the process is adiabatic and quasi-static.

For charging the tank

$$\int (h dm)_{\text{in}} = \Delta U_V = m_2 u_2 - m_1 u_1 \quad (5.34)$$

$$m_p h_p = m_2 u_2 - m_1 u_1$$

where the subscript p refers to the constant state of the fluid in the pipeline. If the tank is initially empty, $m_1 = 0$.

$$m_p h_p = m_2 u_2$$

Since

$$m_p = m_2$$

$$h_p = u_2$$

If the fluid is an ideal gas, the temperature of the gas in the tank after it is charged is given by

$$c_p T_p = c_v T_2$$

or

$$T_2 = \gamma T_p \quad (5.35)$$